

$$0 = 1 - 1 = -1 + 1 = 0$$

From Elementary Arithmetic to Intrinsic Skein-Lasagna

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Your favorite equations?

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Question

Can you prove $0 = 0$ in a nontrivial way?

Test your math level

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$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

imgflip.com



Hopf fibration

Hopf fibration is a map defined by

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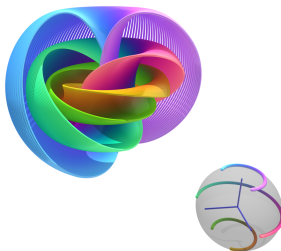
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For any point $x \in \mathbb{S}^2$, preimage $H^{-1}(x)$ is a circle $\mathbb{S}^1$. We can visualize this map by identifying $\mathbb{R}^3 \cong \mathbb{S}^3 - \{\infty\}$, so that the Hopf fibration appears as a map from $\mathbb{R}^3$ to $\mathbb{S}^2$:



Homotopy groups

To understand a space X , we study some homotopy invariants:

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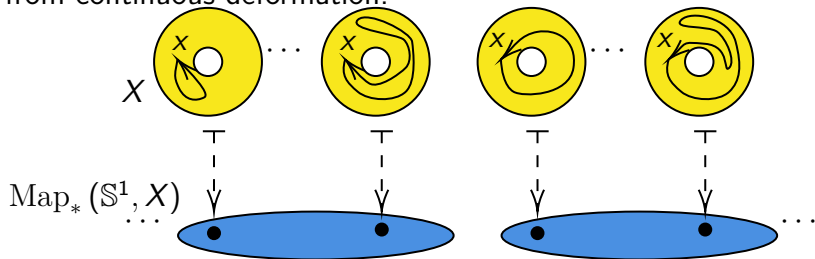
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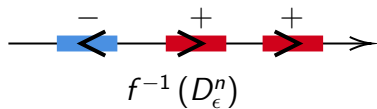
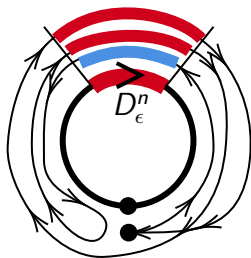
Let us now study the case $X = \mathbb{S}^n$.

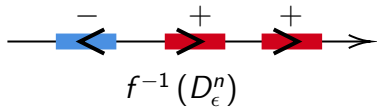
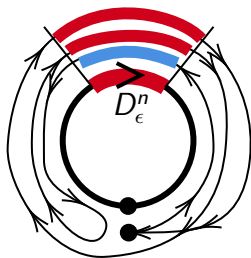
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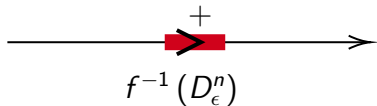
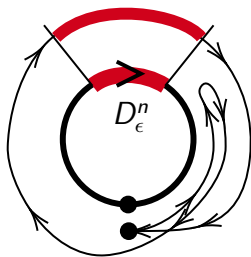
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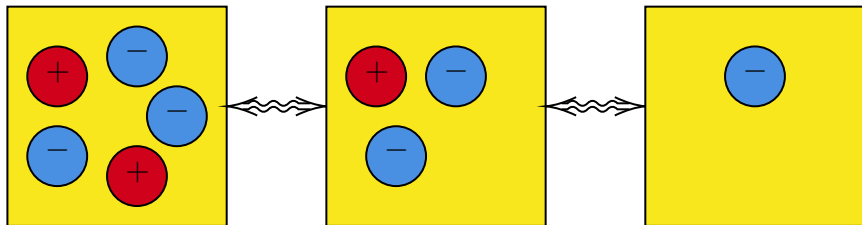
deform



Homotopy groups of spheres

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This gives an isomorphism to the configuration space of disks.

$$\text{End}_*(\mathbb{S}^n) \cong \text{Conf}^{fr}(\mathbb{R}^n) = \bigcup_{j,k} \text{Emb}\left(\bigsqcup_j D_+^n \sqcup \bigsqcup_k D_-^n, \mathbb{R}^n\right)$$

Therefore, $\pi_0 \text{End}_*(\mathbb{S}^n) \cong \mathbb{Z}$, $(j, k) \mapsto j - k$

Higher homotopy groups of spheres

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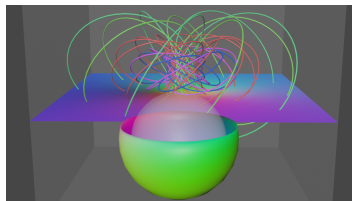
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$$\text{Map}_*(\mathbb{S}^{n+1}, \mathbb{S}^n) \cong \text{Map}_*(\mathbb{S}^1, \text{End}_*(\mathbb{S}^n)) \rightarrow \text{Map}_*([0, 1], \text{End}_*(\mathbb{S}^n))$$

Then H corresponds to a path (that is, a homotopy) from $0 \in \text{End}_*(\mathbb{S}^2)$ back to itself.

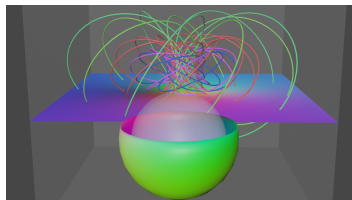
Visualize H as deformation

I made an animation to visualize this deformation.



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We can also visualize it as a movement in the configuration space.
We will show it later.

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We can define the natural numbers $\mathbb{N}$ as the isomorphism classes of finite sets.

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We use the **Grothendieck group** $K_0(\text{FinSet})$:

$$K_0(\text{FinSet}) = \{(X, Y) \in (\text{FinSet}_{/\cong})^2\} / \{(X \sqcup Z, Y \sqcup Z) \sim (X, Y)\}$$

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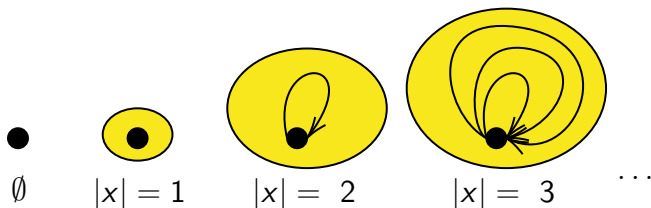
In this way, we may think of (X, Y) as the formal difference $X - Y$.
But what is $K_i(\text{FinSet})$ for $i > 0$?

Space of finite sets

Instead of thinking the set of "finite sets", we now consider the space of them $F = \text{FinSet}^{\cong}$: every point $x \in F$ corresponds to a finite set, and a path between two points corresponds to an isomorphism $f : x \xrightarrow{\cong} y$ of two finite sets.

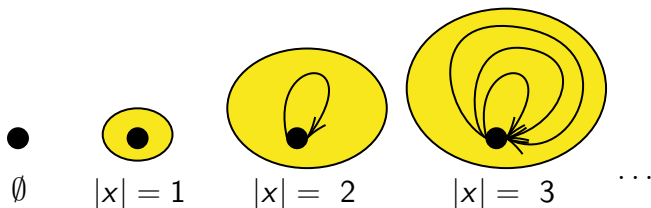
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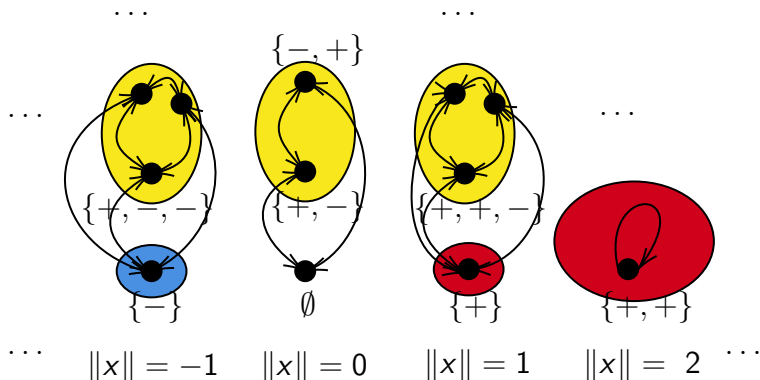
- $\pi_0(F) = \text{FinSet}_{/\cong} \cong \mathbb{N}$.
- $\pi_1(F, x) = S_{|x|}$, the permutation group.
- $\pi_i(F, x) = \{0\}$ for $i > 1$ (one can never deform an isomorphism of set to another).

K-space (spectrum) of finite sets

We now want to add “negative” elements to F . To do this, we consider the space $K(F)$ of ordered finite sets whose elements are marked by $+$ or $-$. We identify signed sets with the same value $\|X\| = |X_+| - |X_-|$, and these identifications are represented by paths.

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Answer

We can prove $0 = 0$ nontrivially with $h : 0 = 1 - 1 = -1 + 1 = 0!$

Algebraic K-groups of a field

Let k be a field. We can carry out the same construction for finite-dimensional k -vector spaces. Let $V_k = \text{FinVect}_k^{\cong}$, and define the **algebraic K-groups** of k by $K_i(k) = \pi_i K(V_k)$. (You may think of finite sets as vector spaces over $\mathbb{F}_1$.)

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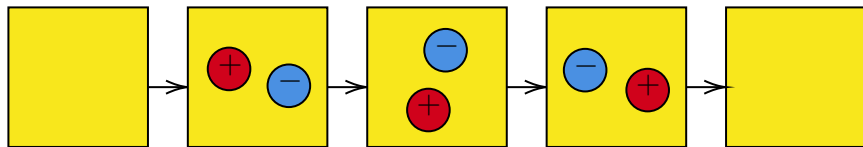
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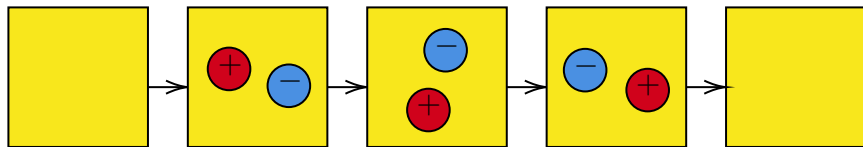


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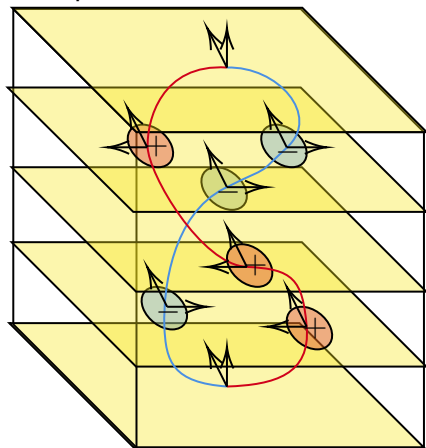


New Question

Can we find a geometric interpretation of the general K -groups of a field $K_i(k)$?

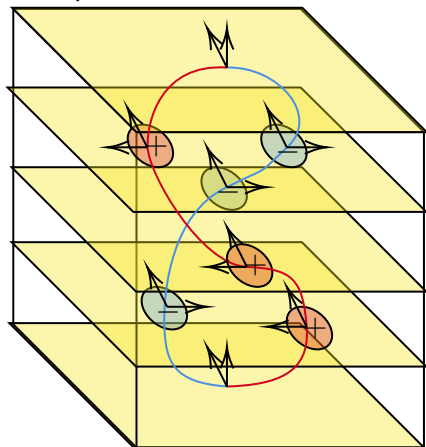
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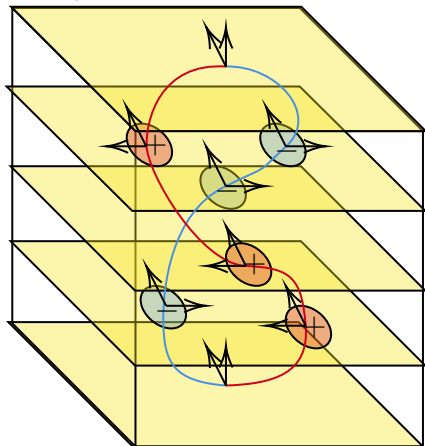
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This gives a framed circle, that is, an embedding $\mathbb{S}^1 \hookrightarrow \mathbb{R}^n$ together with a **trivialization** τ of the normal bundle. We write Ω_i^{fr} for the group of cobordism classes of i -dimensional framed manifolds.

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$$\Omega_i^{fr} \cong \pi_i^s(\mathbb{S})$$

Therefore, this framed circle is nontrivial and generates Ω_1^{fr} .

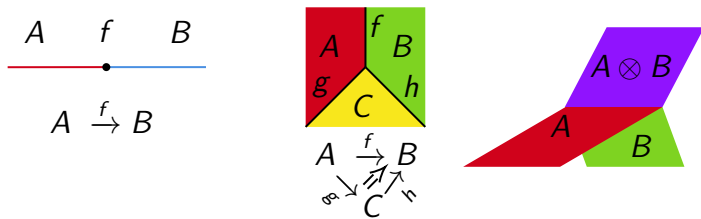
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- 0-codimensional strata are labeled by objects; for $i > 0$, i -codimensional strata are labeled by i -morphisms,
- At the branches of the treefold, we mark how pieces are combined using the monoidal product $\otimes$.



Relation with "Skein-lasagna module"

We do not require an ambient space X into which the lasagna embeds (or, if you prefer, take $X = \mathbb{R}^\infty$). In this sense, we study the intrinsic geometry of the lasagna itself.

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Cobordism of lasagna

As expected, we define a cobordism relation between L_1 and L_2 if there exists a lasagna W with boundary such that $\partial W = L_1 \sqcup -L_2$. Finally, we define the n th $(\mathcal{C}, \otimes)$ -**cobordism group** $\Omega_n^{\mathcal{C}, \otimes}$ to be the abelian group generated by n -lasagnas modulo these cobordism relations.

$(V_k, \oplus)$ -Cobordism Group

As an example, consider $\Omega_n^{V_k, \oplus}$, where $V_k = \text{FinVect}_k^{\cong}$ is the category of finite-dimensional vector spaces with all isomorphisms as morphisms. In this case, one can think of a lasagna as a treefold/manifold equipped with a **k -local system**.

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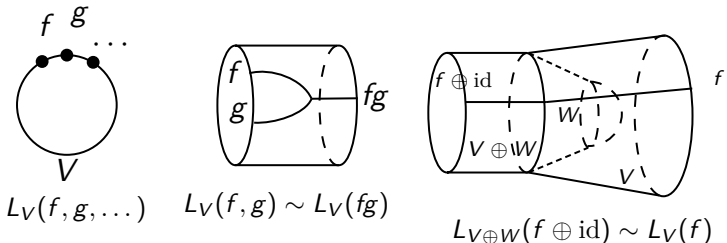
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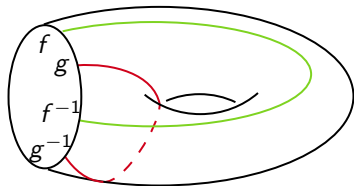
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- Clearly, $\Omega_0^{V_k, \oplus} \cong \mathbb{Z}$, classified by dimension of vector space.
- $\Omega_1^{V_k, \oplus}$ is generated by circles $L_V(f)$ labeled by a vector space V and a matrix $f \in \text{GL}_V(k)$. By stabilizing with identity matrices, we obtain a homomorphism $L : \text{GL}_\infty(k) \twoheadrightarrow \Omega_1^{V_k, \oplus}$.

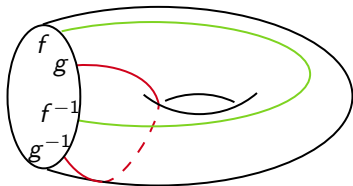


We have one more relation coming from surfaces:



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Notice that $k^\times \cong \mathrm{GL}_\infty(k)/[\mathrm{GL}_\infty(k), \mathrm{GL}_\infty(k)]$. In fact, one can show that the induced homomorphism $L : k^\times \xrightarrow{\cong} \Omega_1^{V_k, \oplus}$ is an isomorphism. The lasagna induced by the Hopf map H is then given by $L(-1)$.

Recall that $K_0(k) = \mathbb{Z}$ and $K_1(k) = k^\times$. This is not just a coincidence:

Generalized Pontryagin–Thom Isomorphism

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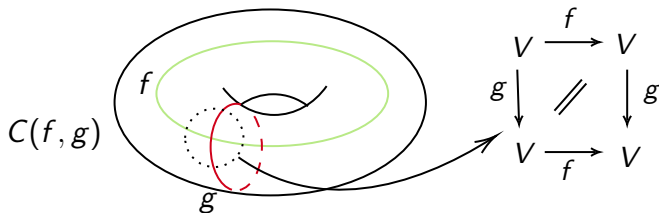
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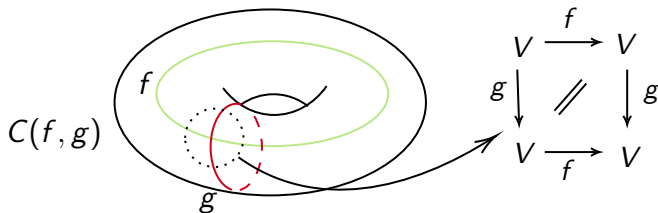
This answers the new question: it suggests that we can view lasagnas as elements of K -groups, providing a bridge from difficult problems in algebra to difficult problems in geometric topology.

So what does this tell us when $n = 2$?

For matrices $f, g \in \mathrm{GL}(k)$ such that $[f, g] = 1$, one can define the following 2-lasagna $C(f, g)$:

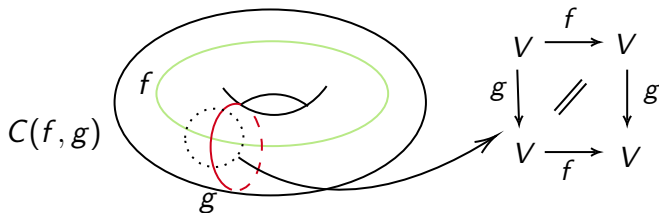


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I do not yet have Ghys' lasagna in hand (it has Heegaard genus less than 26), but I do have an easier example.

3-Lasagna bounds $C(e_{12}(a), e_{12}(b))$

For $i \neq j$, let the elementary matrices $e_{ij}(a)$ be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

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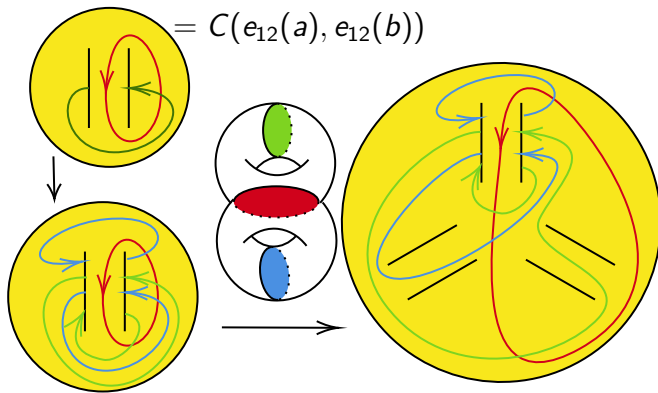
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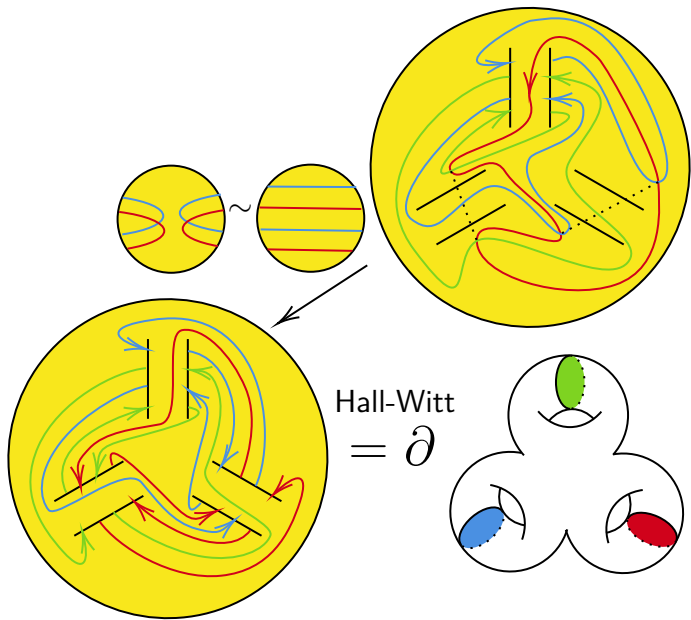
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Using these relations and the Hall–Witt identity, we obtain the following nontrivial proof that $[e_{12}(a), e_{12}(b)] = \mathbf{1}$:

$$\begin{aligned} [e_{12}(a), e_{12}(b)] &= [[e_{13}(a), e_{32}(1)], e_{12}(b)] \\ &= [e_{32}(1), [e_{13}(-a), e_{12}(-b)]]^{e_{13}(a)} [e_{13}(a), [e_{32}(-1), e_{12}(-b)]]^{e_{32}(-1)} \\ &= [e_{32}(1), \mathbf{1}]^{e_{13}(a)} [e_{13}(a), \mathbf{1}]^{e_{32}(-1)} = \mathbf{1} \end{aligned}$$

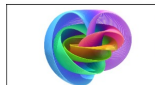
Following this blueprint, we obtain the following Heegaard splitting to construct a lasagna:





Thank you! I hope you enjoyed and gained some new insights.

$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

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